HIC Project 1 Report

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Introduction

The app that we created is Bagl. This is an application that lets users purchase bagels and earn rewards! It also features hotkeys and an order history page so user's and view their previous orders.

We created an application with the goal of fully implementing Schneiderman's 8 golden rules.

8 Golden Rules

Schneiderman's 8 golden rules are:

1. Strive for Consistency
2. Enable Frequent Users to Use Shortcuts
3. Offer Informative Feedback
4. Design Dialogue to Yield Closure
5. Offer Simple Error Handling
6. Permit Easy Reversal of Actions
7. Support Internal Locus of Control
8. Reduce Short Term Memory Load

Best Implementation

The page or feature that best implements each golden rule.

1. All pages, as the design is consistent across the application
2. The Keyboard Shortcuts popup activated when the user presses 'Alt/Option'
3. Order Page
4. Checkout Page
5. Rewards Page
6. Cart Page
7. Cart Page
8. History Page

Home Page

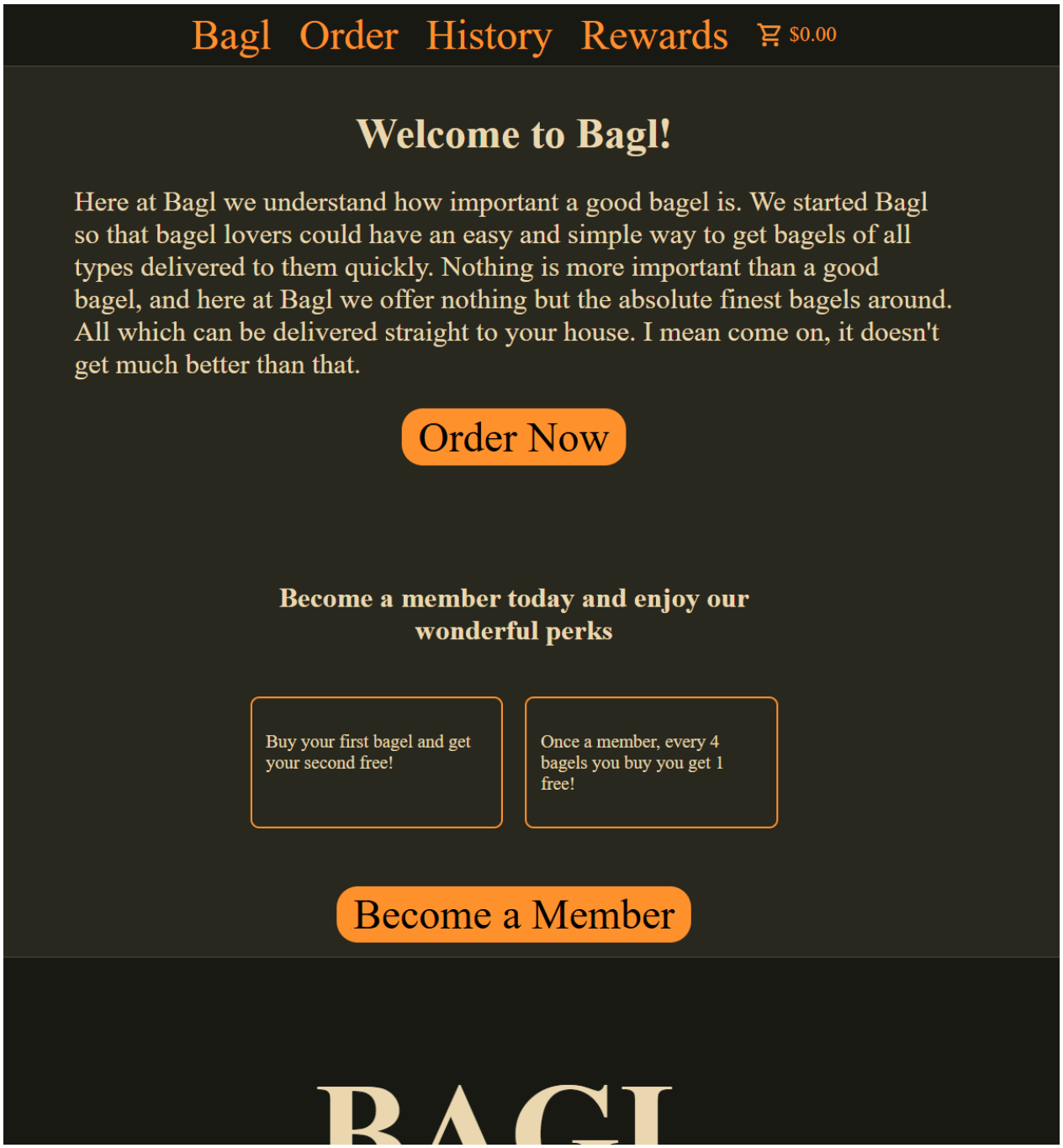


Figure 1: The home page.

When the user first opens the application they are taken to the home page. This page shows the user our rewards and explains the website. It also features buttons that suggest the next actions the user will likely take. The buttons allow the user to quickly place an order or become a member.

The entire application applies the first golden rule of consistency. All of the applications have the same navbar, footer, and color pallette. They share the same accent color of #ff922f orange, our company color. On this home page, the two buttons at the bottom allow the user to be fully in control applying by

allowing them to easily choose to either order now or join the rewards program, applying the seventh golden rule. For keyboard users we also feature a visible outline on all of our navbar buttons and also our ‘Order Now’ and ‘Become a Member’ buttons, this applies the 2nd golden rule of universal usability.

Order Page

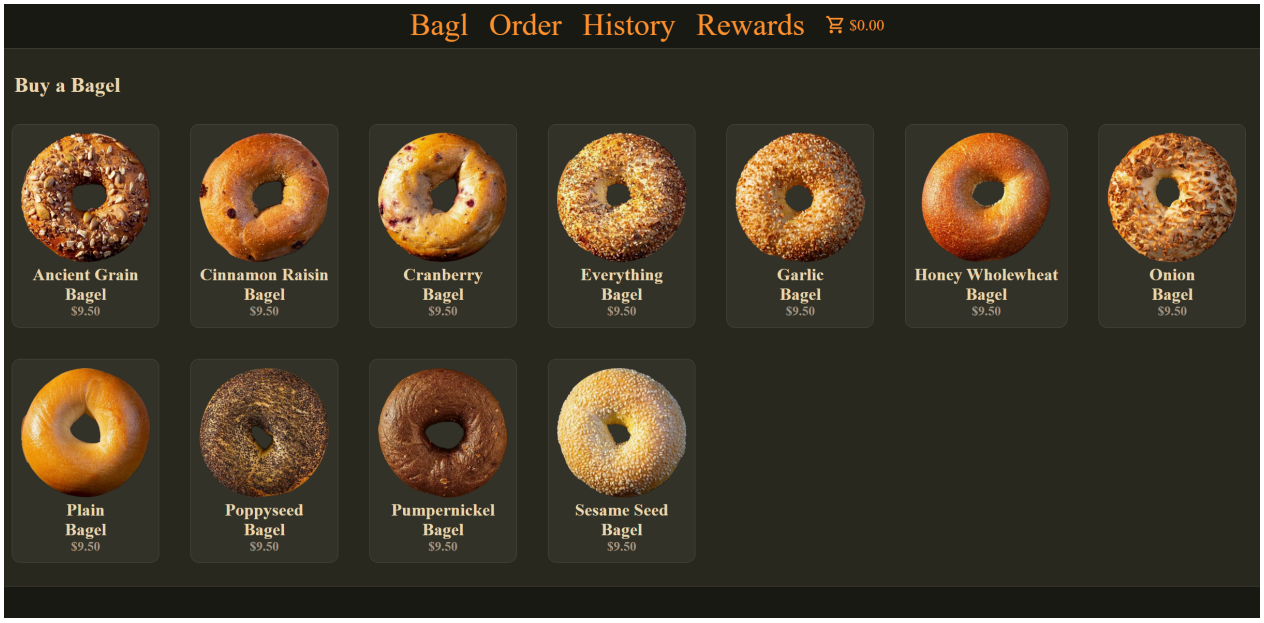


Figure 2: The order page.

The order pag shows all of our different bagel options. The user can easily click on any of the bagels to add one of them to their cart. Once they click on one of the bagels, they are taken to the Bagel Info page to increase or decrease their bagel count.

This page implementes golden rule eight as it has the bagel image, name, and price on every card, reducing the short term memory load needed by the user. It also pops out each bagel, and highlights on hover, as informative feedback for the user implementing golden rule three.

Bagel Info Page



Figure 3: The bagel info page.

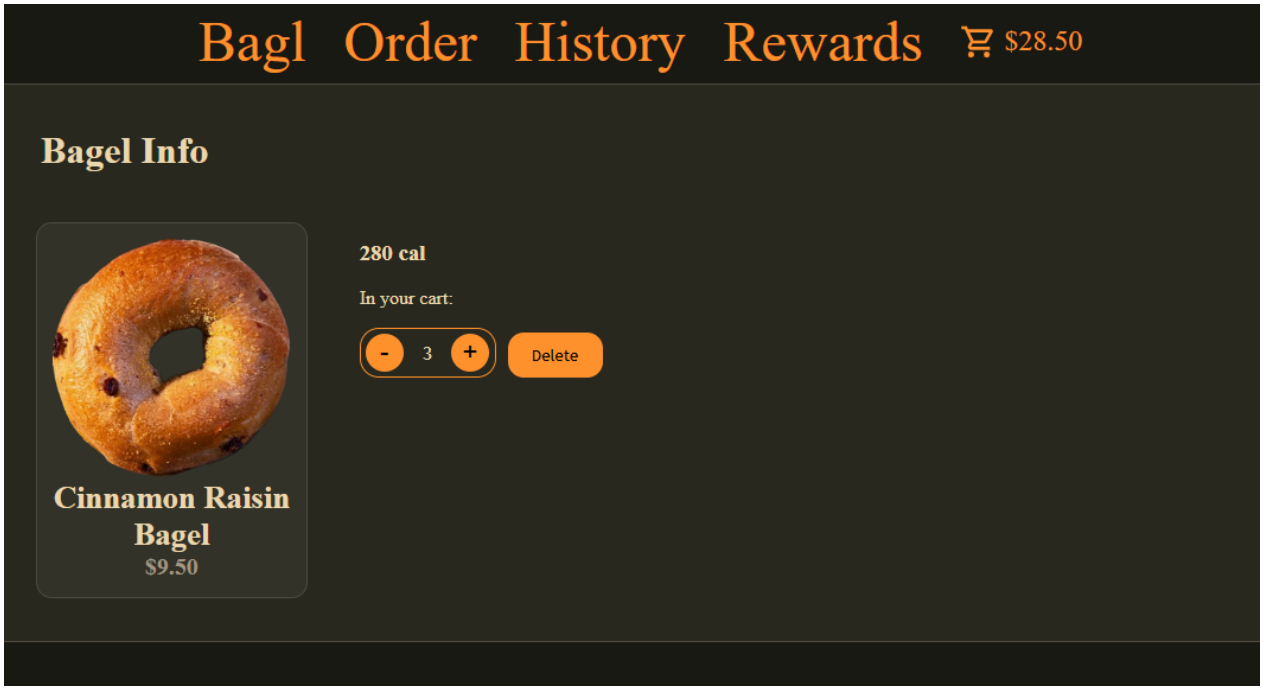


Figure 4: The bagel info page after pressing 'Add to Cart'.

After clicking on a bagel the user is taken to the Bagel Info Page. This allows the user to see the calorie information relating to each bagel and allows them to confirm that they would like to add the bagel to the cart. Once the bagel is added to the cart the user can then increase or decrease the number of bagels that they would like to purchase. If they decide against purchasing the bagel they can easily press the

`Delete` button. The cost in the cart also updates live as the number of bagels in the cart increases and decreases.

This page offers informative feedback as pressing add to cart quickly becomes the quantity controls. The cost of the entire purchase also updating in the navbar as the user adds and subtracts amounts adds to this informative feedback implementing the third golden rule. We offer simple error handling as we disable the minus button at 1 bagel. This prevents the user from having 0 or negative bagels, implementing rule five. We also allow the user to undo their actions as `Delete` removes the bagel from the cart and the minus button allows the user to undo accidentally adding an extra bagel. This implements rule six. We also reduce short term memory load as the user can see exactly how many of that bagel are in their cart in between the plus and minus, and the total cost of the transaction easily in the top right. This reduction of short term memory load helps to implement rule eight.

Cart Page



Figure 5: The cart page.

The cart page allows the user to see what bagels they are preparing to purchase. It allows us as users to increase and decrease count of each bagel. The checkout page also has a `Checkout` button that when clicked makes the purchase. When the user has rewards saved up, they can spend these rewards also on the checkout page by checking a box to confirm that they want to use the reward to get a free bagel.

The order summary always shows the item count and the subtotal which reduces the short term memory load of the user allowing them to always know exactly how much the purchase will cost them even as they add or remove bagels, implementing rule eight. The checkout is disabled when the cart is empty and a message is shown that tells the user their cart is empty. This helps to prevent errors before they happen and implements rule five. We are also able to easily increase the counts of our items and

derease them easily, like the bagel info page, allowing easy reversal of actions, implementing rule six. By allowing the user to easily decide when to checkout, modify bagel counts, and have their total on the side, we make the user feel like they are completely in control implementing rule seven. By having the same quantity controls as the bagel information page we establish consistency across the application.

Checkout Page

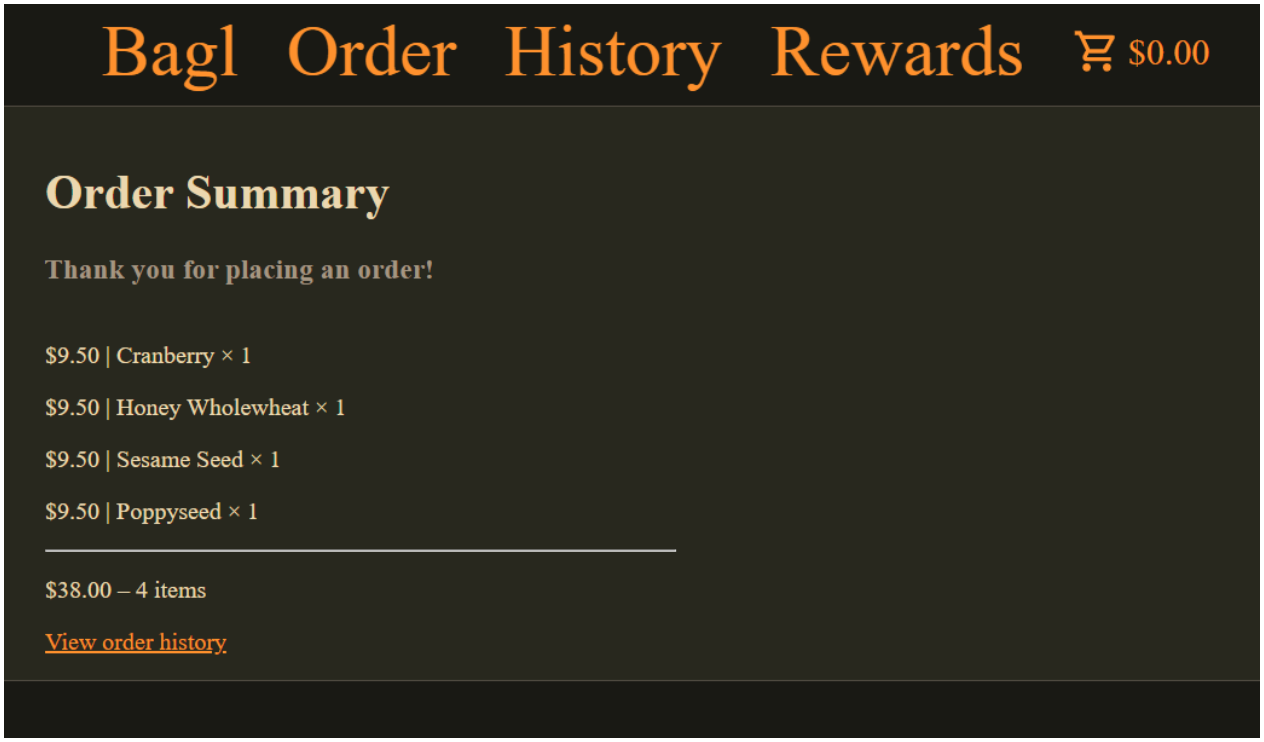


Figure 6: The checkout page.

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The checkout page has a thank you message and itemized receipt after completing the purchase. This dialogue yields closure to the user that their purchase went through implementing golden rule four. The `Checkout` button is also grayed out in the order summary preventing errors before they happen and implementing golden rule five.

History Page

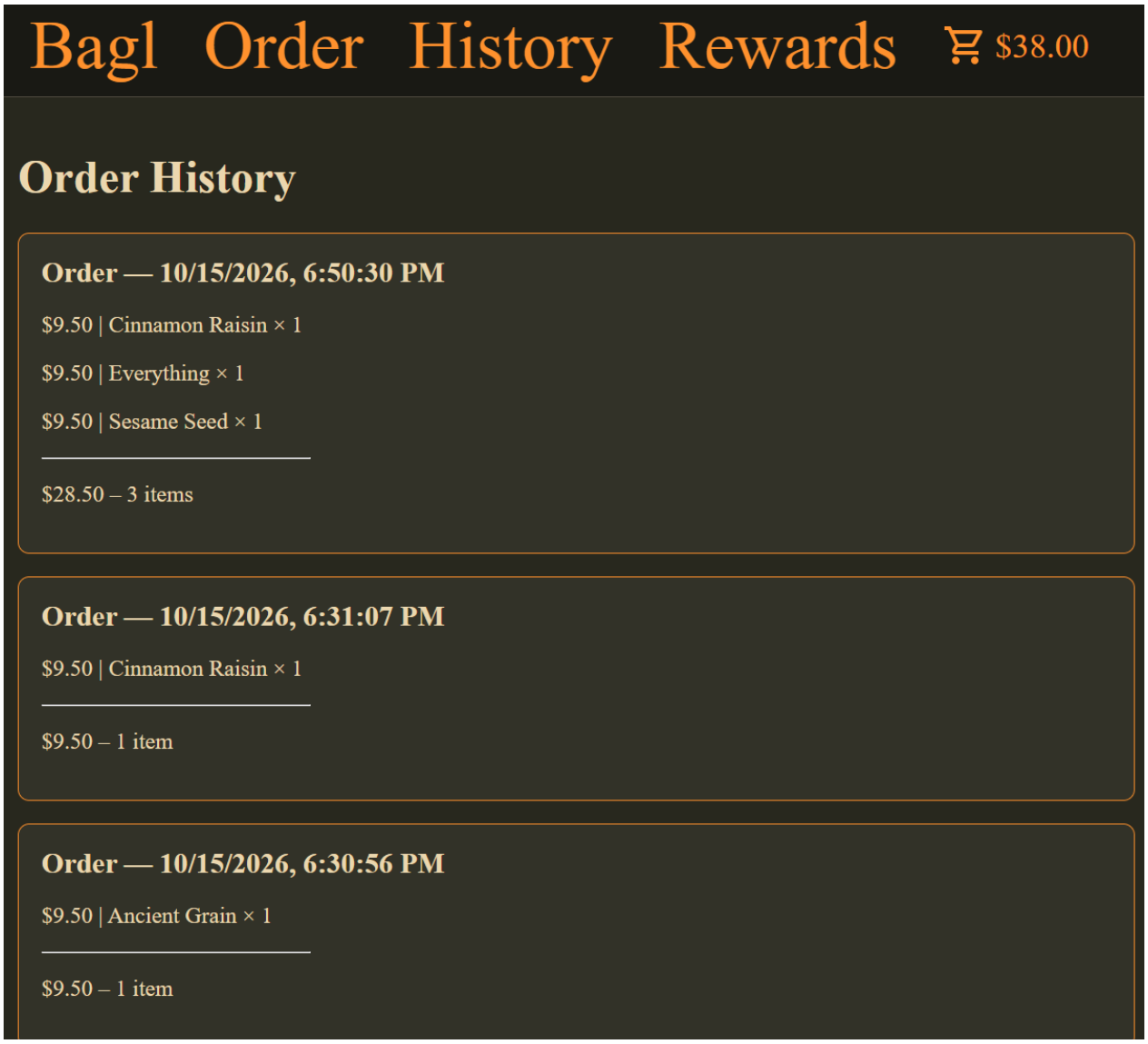


Figure 7: The history page.

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Rewards Page

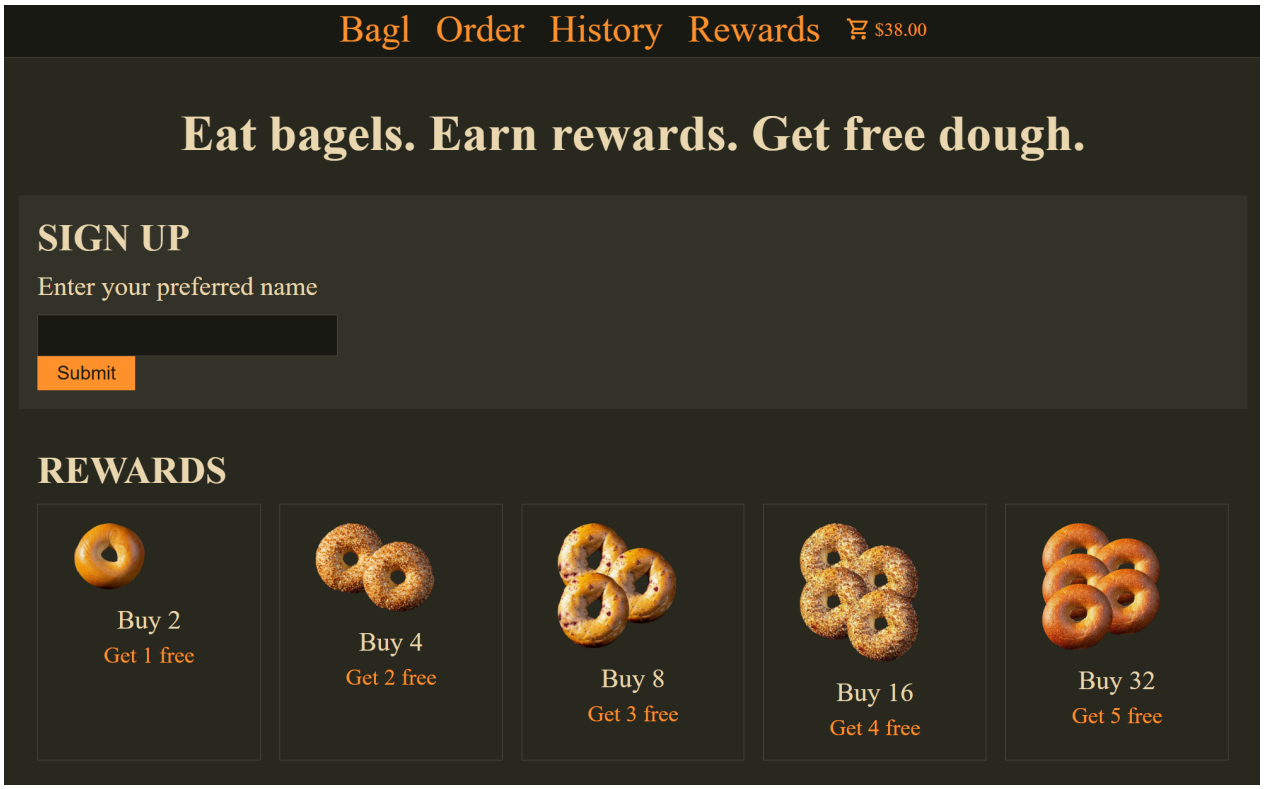


Figure 8: The rewards page.

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Hotkeys



Figure 9: The hotkeys dropdown acccessed by pressing Alt/Option.

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Group Contributions

Running the application

You can check out the application at: https://bagl-pi.vercel.app/

If you would like to run the application locally, First run

```
npm install
```

Then to run the application use

```
npm run dev
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